

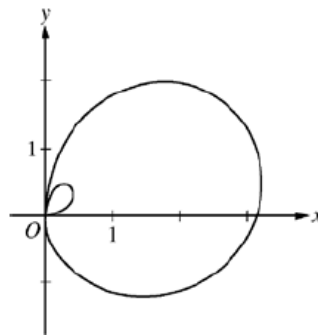
“Hopefully you did them.” Me

FR1. 2014 Practice Exam (Calculator)

Consider the polar curve defined by the function $r(\theta) = \theta \cos \theta$, where $0 \leq \theta \leq \frac{3\pi}{2}$.

The derivative of r is given by $\frac{dr}{d\theta} = \cos \theta - \theta \sin \theta$. The figure above shows

the graph of r for $0 \leq \theta \leq \frac{3\pi}{2}$.

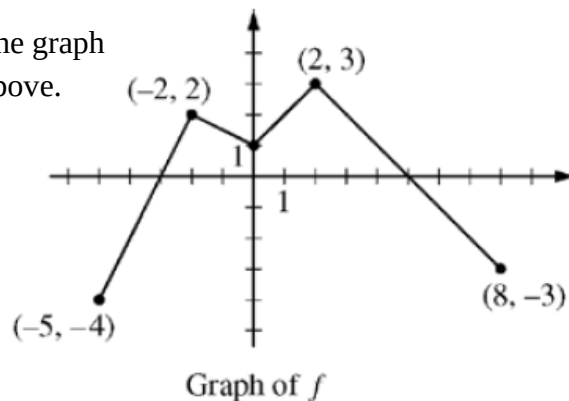


- Find the area of the region enclosed by the inner loop of the curve.
- For $0 \leq \theta \leq \frac{3\pi}{2}$, find the greatest distance from any point on the graph of r to the origin. Justify your answer.
- There is a point on the curve at which the slope of the line tangent to the curve is $\frac{2}{2-\pi}$. At this point, $\frac{dy}{d\theta} = \frac{1}{2}$. Find $\frac{dx}{d\theta}$ at this point.

FR2. 2014 Practice Exam (No Calculator)

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above.

Let g be the function given by $g(x) = 3x + \int_{-2}^x f(t) dt$.



- Find $g(0)$ and $g(-5)$.
- Find $g'(x)$ in terms of $f(x)$. For each of $g''(-2)$ and $g''(4)$, Find the value or state that it does not exist.
- On what intervals, if any, is the graph of g concave down? Give a reason for your answer.
- The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

MC1. For $x > 0$, $\frac{d}{dx} \int_1^{\sqrt{x}} \frac{1}{1+t^2} dt =$

- (A) $\frac{1}{2\sqrt{x}(1+x)}$ (B) $\frac{1}{2\sqrt{x}(1+\sqrt{x})}$ (C) $\frac{1}{1+x}$ (D) $\frac{\sqrt{x}}{1+x}$

MC2. Which of the following series converge?

I. $1 + (-1) + 1 + \cdots + (-1)^{n-1} + \cdots$

II. $1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} + \cdots$

III. $1 + \frac{1}{3} + \frac{1}{3^2} + \cdots + \frac{1}{3^{n-1}} + \cdots$

- (A) I only
(B) II only
(C) III only
(D) II and III only

MC3. Let f be the function defined by $f(x) = \begin{cases} x^2 + 2 & \text{for } x \leq 3, \\ 6x + k & \text{for } x > 3. \end{cases}$

If f is continuous at $x = 3$, what is the value of k ?

- (A) -7 (B) 2 (C) 3 (D) 7 (E) There is no such value of k .

MC4. What is the sum of the series $\sum_{n=1}^{\infty} \frac{(-2)^n}{e^{n+1}}$?

- (A) $\frac{-2}{e^2 - 2e}$ (B) $\frac{-2}{e^2 + 2e}$ (C) $\frac{-2}{e + 2}$ (D) $\frac{e}{e + 2}$